



Contact



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The Hong Kong University of Science
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Clear Water Bay, Kowloon
Hong Kong

Education

Ph.D. in Environmental Engineering
(Expected Aug 2030)
The Hong Kong University of Science
and Technology

**M.S. in Environmental Science &
Engineering**
(Jun 2025)
Fudan University, China
GPA: 3.4/4.0

B.S. in Environmental Engineering
(Jun 2022)
Fudan University, China
GPA: 3.2/4.0

Skills & Expertise

- Anaerobic and microaerobic reactor operation
- Microbial cultivation and fermentation
- Shotgun metagenomic and metatranscriptomic analysis
- Bioinformatic analysis of microbial and viral metagenomes
- Virus–host interaction and ecological network analysis

Curriculum Vitae

Keliang Zhu

PhD student in Environmental Engineering at the Hong Kong University of Science and Technology (HKUST), interested in anaerobic digestion and microbial ecology for sustainable waste treatment and resource recovery. His current research focuses on microaeration-assisted anaerobic digestion, aiming to understand how microaeration reshapes microbial communities and interactions and how these changes can be harnessed to improve anaerobic digestion performance.

Professional Experience

- **Aug 2026–Current**
Graduate Research Assistant
Department of Civil and Environmental Engineering
The Hong Kong University of Science and Technology,
Hong Kong
Research focus: Microbial communities, functions, and interactions under microaeration in anaerobic digestion
PI: Prof. Samir Kumar Khanal
- **Sep 2022–Dec 2025**
Graduate Research Assistant
Department of Environmental Science and Engineering
Fudan University, China
Research focus: Viral communities, functions, and virus–host interactions in anaerobic digestion
Supervisor: Prof. Gang Luo
- **Sep 2021–Jun 2022**
Undergraduate Research Assistant
Department of Environmental Science and Engineering
Fudan University, China
Research focus: Biological production of PHA from phenol using mixed microbial cultures
Supervisor: Prof. Yan Zhang

Research Areas

- Microbial communities, functional responses, and quorum sensing in microaerobic anaerobic digestion
- Occurrence and dissemination risks of antibiotic resistance genes in microaerobic anaerobic digestion
- Viral communities in hydrochar-enhanced anaerobic digestion
- Viral communities, functions, and virus–host interactions in anaerobic digestion exposed to PS and PVC microplastics

Publications

1. **Zhu, K.**, Shi, Z., Zhang, Y., Zhang, S., Luo, G. (2025). Unravelling DNA viral communities involved in the anaerobic digestion of waste activated sludge enhanced by hydrochar. *Journal of Environmental Management*, 392, 126761. <https://doi.org/10.1016/j.jenvman.2025.126761>
2. Shao, Q., Luo, F., Zhang, Z., Zhang, C., **Zhu, K.**, Zhang, Y., Zhang, S., Luo, G. (2026). From Promotion to Inhibition: A Holistic Perspective on the Biological Effects of Polylactic Acid Microplastics on Black Soldier Fly Larva in Food Waste Bioconversion. *Environmental Science&Engineering*, 60(1):350-363. <https://doi.org/10.1021/acs.est.5c16712>
3. Zou, Y., Yang, M., Tao, Q., **Zhu, K.**, Liu, X., Wan, C., Harder, M. K., Yan, Q., Liang, B., Ntaikou, I., Antonopoulou, G., Lyberatos, G., & Zhang, Y. (2023). Recovery of polyhydroxyalkanoates (PHAs) polymers from a mixed microbial culture through combined ultrasonic disruption and alkaline digestion. *Journal of Environmental Management*, 326, 116786. doi.org/10.1016/j.jenvman.2022.116786
4. Shao, Q., Sun, J., **Zhu, K.**, Ngo, H. H., Zhang, Y., Zhang, C., Zhang, S., & Luo, G. (2025). Enhanced reactor stability to shock load with hydrochar: Responses of anaerobic granular sludge. *Chemical Engineering Journal*, 503, 158274. doi.org/10.1016/j.cej.2024.158274
5. Shao, Q., Luo, T., Zhang, C., **Zhu, K.**, Zhang, S., & Luo, G. (2025). Enhanced anaerobic phenol degradation: Critical roles of glucose and hydrochar on microbial traits. *Bioresource Technology*, 429, 132490. doi.org/10.1016/j.biortech.2025.132490
6. Shi, Y., Chen, Z., **Zhu, K.**, Fan, J., Clark, J. H., Luo, G., & Zhang, S. (2022). Speciation evolution and transformation mechanism of P during microwave hydrothermal process of sewage sludge. *Science of the Total Environment*, 815, 152801. doi.org/10.1016/j.scitotenv.2021.152801

Awards and Honors

1. **PostGraduate Studentship**–The Hong Kong University of Science and Technology–Feb 2026–Current
2. **Academic Excellence Scholarship**–Fudan University–Sep 2022–Jun 2025

Relevant Courses

Graduate Courses

(M.S. in Environmental Science & Engineering)

- Principles of Environmental Chemistry
- Foundation of Environmental Engineering
- Environmental Assessment
- Industrial Water Pollution Control
- Biomass energy technologies
- The frontiers of technologies for resource recovery and high value added products production from organic wastes
- Biochar and Environment

Undergraduate Main Courses

(B.S. in Environmental Engineering)

- Ecology
- Environmental Biology
- Organic Chemistry
- Environmental Analytical Chemistry
- Environmental Physical Chemistry
- Environmental Chemistry
- Principles and Methods of Air Pollution Control
- Water Pollution and Control
- Solid Waste Treatment and Recycling
- Probability and Environmental Statistics
- Industrial Wastewater Treatment
- Soil Environmental Chemistry
- Environmental Management
- Environmental Assessment